

Measure Theory – 6

Conditional Expectation and the Radon–Nikodym Theorem,
Symmetry Arguments, Bayesian Posterior Densities,
and the Weak Law of Large Numbers with Slutsky’s Theorem

Ravleen Bajaj

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, λ is Lebesgue measure, and $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra.

1 Conditional Expectation and Radon–Nikodym Theory

1.1 Definition, Existence, and Uniqueness

Definition 1.1 (Conditional expectation). Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. The **conditional expectation** $\mathbb{E}[X \mid \mathcal{G}]$ is any random variable Z such that

- (i) Z is $\mathcal{G}$ -measurable, and
- (ii) $\int_G X \, d\mathbb{P} = \int_G Z \, d\mathbb{P}$ for every $G \in \mathcal{G}$.

The **conditional probability** of an event A is $\mathbb{P}(A \mid \mathcal{G}) = \mathbb{E}[\mathbf{1}_A \mid \mathcal{G}]$, and $\mathbb{E}[X \mid Y] := \mathbb{E}[X \mid \sigma(Y)]$.

Theorem 1.2 (Radon–Nikodym). Let μ, ν be σ -finite measures on $(\Omega, \mathcal{G})$ with $\nu \ll \mu$ (that is, $\mu(G) = 0 \Rightarrow \nu(G) = 0$). Then there is a $\mathcal{G}$ -measurable $f \geq 0$, unique μ -a.e., with

$$\nu(G) = \int_G f \, d\mu \quad (G \in \mathcal{G}).$$

The function $f = \frac{d\nu}{d\mu}$ is called the **Radon–Nikodym derivative**.

Theorem 1.3 (Existence and uniqueness of $\mathbb{E}[X \mid \mathcal{G}]$). For every $X \in L^1$, $\mathbb{E}[X \mid \mathcal{G}]$ exists and is unique up to $\mathbb{P}$ -null sets; it is integrable.

Proof. Existence. Let first $X \geq 0$ and define $\nu(G) = \int_G X \, d\mathbb{P}$ on $(\Omega, \mathcal{G})$. This is a finite measure, and $\nu \ll \mathbb{P}|_{\mathcal{G}}$. Radon–Nikodym gives a $\mathcal{G}$ -measurable $Z = d\nu/d\mathbb{P}|_{\mathcal{G}} \geq 0$ with $\int_G Z \, d\mathbb{P} = \nu(G) = \int_G X \, d\mathbb{P}$. For general X apply this to X^+ and X^- and subtract; both are finite because $X \in L^1$. *Uniqueness.* If Z, Z' both satisfy (i)–(ii), then $G = \{Z > Z'\} \in \mathcal{G}$ and $\int_G (Z - Z') \, d\mathbb{P} = 0$ with integrand > 0 on G , so $\mathbb{P}(G) = 0$; symmetrically for $\{Z < Z'\}$. $\square$

Reading the definition.

$\mathbb{E}[X \mid \mathcal{G}]$ is the $\mathcal{G}$ -measurable “best summary” of X : it has the same integral as X over every set

that $\mathcal{G}$ can see. It is a random variable defined only up to null sets, so every identity involving it holds almost surely, and any version is called a *version*.

1.2 Properties

Proposition 1.4 (Properties of conditional expectation). *Let $X, Y \in L^1$ and let $\mathcal{G}, \mathcal{H}$ be sub- σ -algebras. Almost surely:*

- (a) **Linearity:** $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$.
- (b) **Total expectation:** $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$.
- (c) **Monotonicity:** $X \leq Y \implies \mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}]$.
- (d) **Tower property:** if $\mathcal{H} \subseteq \mathcal{G}$, then $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$.
- (e) **Taking out what is known:** if W is $\mathcal{G}$ -measurable and $WX \in L^1$, then $\mathbb{E}[WX \mid \mathcal{G}] = W\mathbb{E}[X \mid \mathcal{G}]$.
- (f) **Independence:** if X is independent of $\mathcal{G}$, then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.
- (g) **Measurable case:** if X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$.
- (h) **Jensen:** for convex φ with $\varphi(X) \in L^1$, $\varphi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\varphi(X) \mid \mathcal{G}]$.

Proof. Each is verified by checking (i)–(ii) for the proposed right-hand side and invoking uniqueness. (a) The right side is $\mathcal{G}$ -measurable and integrates correctly over each $G \in \mathcal{G}$ by linearity of the integral. (b) Take $G = \Omega$. (c) With $G = \{\mathbb{E}[X \mid \mathcal{G}] > \mathbb{E}[Y \mid \mathcal{G}]\} \in \mathcal{G}$, $\int_G (\mathbb{E}[X \mid \mathcal{G}] - \mathbb{E}[Y \mid \mathcal{G}]) = \int_G (X - Y) \leq 0$ while the integrand is positive on G , so $\mathbb{P}(G) = 0$. (d) For $H \in \mathcal{H} \subseteq \mathcal{G}$, $\int_H \mathbb{E}[X \mid \mathcal{G}] = \int_H X$, and $\mathbb{E}[X \mid \mathcal{H}]$ is the unique $\mathcal{H}$ -measurable function with this property. (e) Ladder: for $W = \mathbf{1}_{G_0}$, $\int_G \mathbf{1}_{G_0} X = \int_{G \cap G_0} X = \int_{G \cap G_0} \mathbb{E}[X \mid \mathcal{G}] = \int_G \mathbf{1}_{G_0} \mathbb{E}[X \mid \mathcal{G}]$; extend to simple W by linearity, to $W \geq 0$ by MCT, and to general W by DCT. (f) $\mathbb{E}[X]$ is constant, hence $\mathcal{G}$ -measurable, and $\int_G X = \mathbb{E}[X \mathbf{1}_G] = \mathbb{E}[X] \mathbb{P}(G) = \int_G \mathbb{E}[X]$ by independence. (g) is immediate from the definition. (h) follows from writing φ as a supremum of countably many affine functions. $\square$

Proposition 1.5 (Conditioning on a countable partition). *If $\mathcal{G} = \sigma(\{A_i\}_{i \in I})$ for a countable partition with $\mathbb{P}(A_i) > 0$, then*

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{i \in I} \frac{\mathbb{E}[X \mathbf{1}_{A_i}]}{\mathbb{P}(A_i)} \mathbf{1}_{A_i}.$$

Proof. The right side is constant on each A_i , hence $\mathcal{G}$ -measurable, and every $G \in \mathcal{G}$ is a union of atoms, over which both sides integrate to $\sum_{i: A_i \subseteq G} \mathbb{E}[X \mathbf{1}_{A_i}]$. $\square$

Proposition 1.6 (L^2 -projection). *If $X \in L^2$, then $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace $L^2(\Omega, \mathcal{G}, \mathbb{P})$: for every $\mathcal{G}$ -measurable $W \in L^2$,*

$$\mathbb{E}[(X - W)^2] = \mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])^2] + \mathbb{E}[(\mathbb{E}[X \mid \mathcal{G}] - W)^2].$$

Hence $\mathbb{E}[X \mid \mathcal{G}]$ minimizes the mean squared error among $\mathcal{G}$ -measurable predictors.

Proof. Write $Z = \mathbb{E}[X \mid \mathcal{G}]$. By property (e), $\mathbb{E}[(X - Z)(Z - W)] = \mathbb{E}[(Z - W) \mathbb{E}[X - Z \mid \mathcal{G}]] = 0$, so expanding $(X - W)^2 = ((X - Z) + (Z - W))^2$ leaves no cross term. $\square$

Remark. By the Doob–Dynkin lemma, $\mathbb{E}[X | Y] = \varphi(Y)$ for some Borel function φ ; one writes $\varphi(y) = \mathbb{E}[X | Y = y]$. Only $\varphi(Y)$ (a random variable, unique a.s.) is canonical; φ itself is determined only $\mathbb{P}_Y$ -a.e.

Note:

- $\mathbb{E}[X | \mathcal{G}]$ is a *random variable*, not a number, and is defined only a.s.
- Property (e) requires W to be $\mathcal{G}$ -measurable; you cannot pull out a factor that is not.
- $\mathbb{E}[X | \mathcal{G}] = \mathbb{E}[X]$ requires independence of X and $\mathcal{G}$, not merely zero correlation.

1.3 Problem: Conditioning on a Non-Injective Function

On $\Omega = (0, 1]$ with $\mathbb{P} = \lambda$, let

$$Y(x) = \min\left(x, \frac{1}{2}\right).$$

The σ -algebra $\sigma(Y)$. For a Borel set B ,

$$Y^{-1}(B) = \left(B \cap (0, \tfrac{1}{2})\right) \cup \begin{cases} [\tfrac{1}{2}, 1] & \text{if } \tfrac{1}{2} \in B, \\ \emptyset & \text{otherwise.} \end{cases}$$

Hence

$$\sigma(Y) = \left\{ C, \ C \cup [\tfrac{1}{2}, 1] : C \subseteq (0, \tfrac{1}{2}) \text{ Borel} \right\}.$$

Its atoms are the singletons $\{y\}$ for $y \in (0, \frac{1}{2})$ and the single big atom $[\frac{1}{2}, 1]$, on which Y is constant. (The assignment writes this last atom as $(\frac{1}{2}, 1]$; the point $\frac{1}{2}$ is a null set, and it cannot be separated from $(\frac{1}{2}, 1]$ by $\sigma(Y)$, since $Y^{-1}(\{\frac{1}{2}\}) = [\frac{1}{2}, 1]$.)

Claim. For $X \in L^1$,

$$\mathbb{E}[X | Y](x) = X(x) \mathbf{1}_{(0, 1/2)}(x) + 2 \left(\int_{[1/2, 1]} X \, d\lambda \right) \mathbf{1}_{[1/2, 1]}(x).$$

In particular, for an event A ,

$$\mathbb{P}(A | Y)(x) = \begin{cases} \mathbf{1}_A(x), & x \in (0, \tfrac{1}{2}), \\ 2 \lambda(A \cap [\tfrac{1}{2}, 1]), & x \in [\tfrac{1}{2}, 1]. \end{cases}$$

Proof. Measurability. On $(0, \frac{1}{2})$ the σ -algebra $\sigma(Y)$ restricted to $(0, \frac{1}{2})$ is the whole Borel σ -algebra, so $X \mathbf{1}_{(0, 1/2)}$ is $\sigma(Y)$ -measurable; the second term is constant on the atom $[\frac{1}{2}, 1]$. *Defining property.* Let $G = C$ or $G = C \cup [\frac{1}{2}, 1]$ with $C \subseteq (0, \frac{1}{2})$ Borel, and call the right side Z . Then

$$\int_C Z \, d\lambda = \int_C X \, d\lambda, \quad \int_{[1/2, 1]} Z \, d\lambda = \tfrac{1}{2} \cdot 2 \int_{[1/2, 1]} X \, d\lambda = \int_{[1/2, 1]} X \, d\lambda.$$

Adding gives $\int_G Z = \int_G X$ in either case. $\square$

Example 1.7. Take $A = (\frac{1}{4}, \frac{3}{4}]$. Then $\mathbb{P}(A | Y) = \mathbf{1}_{(1/4, 1/2)}$ on $(0, \frac{1}{2})$, and on $[\frac{1}{2}, 1]$ it equals $2 \cdot \frac{1}{4} = \frac{1}{2}$. As a check, $\mathbb{E}[\mathbb{P}(A | Y)] = \lambda((\frac{1}{4}, \frac{1}{2})) + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = \lambda(A)$.

Remark. The example is “non-symmetric” in that Y separates points on $(0, \frac{1}{2})$ (so conditioning there leaves X unchanged) but collapses $[\frac{1}{2}, 1]$ to a single value (so conditioning there averages X). The conditional expectation is the identity where information is complete and a local average where it is not.

2 Symmetry and Conditional Expectation Calculations

2.1 Symmetry Principle

Proposition 2.1 (Exchangeable summands). *Let $X_1, \dots, X_m$ be i.i.d. and integrable, and $S_m = X_1 + \dots + X_m$. Then*

$$\mathbb{E}[X_1 | S_m] = \dots = \mathbb{E}[X_m | S_m] = \frac{S_m}{m} \quad a.s.$$

Proof. Fix i and a Borel set B . Since $(X_1, \dots, X_m)$ and $(X_i, X_2, \dots, X_1, \dots)$ (with X_1, X_i swapped) have the same joint law (exchangeability of i.i.d. vectors) and S_m is invariant under the swap,

$$\mathbb{E}[X_1 \mathbf{1}_{\{S_m \in B\}}] = \mathbb{E}[X_i \mathbf{1}_{\{S_m \in B\}}].$$

As every set in $\sigma(S_m)$ has the form $\{S_m \in B\}$, this shows $\mathbb{E}[X_1 | S_m] = \mathbb{E}[X_i | S_m]$ a.s. Summing over i and using linearity and $\mathbb{E}[S_m | S_m] = S_m$ (measurable case) gives $m \mathbb{E}[X_1 | S_m] = S_m$. $\square$

2.2 Problem: Two Exponentials

Let $X_1, X_2 \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1)$ and $S = X_1 + X_2$. **Claim:** $\mathbb{E}[X_1 | S] = S/2$.

Proof. By linearity,

$$\mathbb{E}[X_1 | S] + \mathbb{E}[X_2 | S] = \mathbb{E}[X_1 + X_2 | S] = \mathbb{E}[S | S] = S.$$

By Proposition 2.1, $\mathbb{E}[X_1 | S] = \mathbb{E}[X_2 | S]$, so each equals $S/2 = (X_1 + X_2)/2$. $\square$

Consistency check via the conditional law. The map $(x_1, x_2) \mapsto (x_1, s) = (x_1, x_1 + x_2)$ has Jacobian 1, so the joint density of (X_1, S) is

$$f(x, s) = e^{-x} e^{-(s-x)} = e^{-s}, \quad 0 < x < s,$$

and $S \sim \text{Gamma}(2, 1)$ has density se^{-s} . Hence the conditional density is

$$f_{X_1|S}(x | s) = \frac{e^{-s}}{se^{-s}} = \frac{1}{s}, \quad 0 < x < s,$$

that is, $X_1 | S = s \sim \text{Uniform}(0, s)$. This gives

$$\mathbb{E}[X_1 | S] = \frac{S}{2}, \quad \mathbb{E}[X_1^2 | S] = \frac{S^2}{3}, \quad \text{Var}(X_1 | S) = \frac{S^2}{12}.$$

The law of total variance confirms consistency:

$$\mathbb{E}[\text{Var}(X_1 | S)] + \text{Var}(\mathbb{E}[X_1 | S]) = \frac{\mathbb{E}[S^2]}{12} + \frac{\text{Var}(S)}{4} = \frac{6}{12} + \frac{2}{4} = 1 = \text{Var}(X_1).$$

Remark. The symmetry argument uses only that the X_i are i.i.d. and integrable; the exponential distribution enters only in the finer statement that the conditional law is uniform on $(0, S)$.

Note:

- Symmetry gives $\mathbb{E}[X_1 | S] = \mathbb{E}[X_2 | S]$ only because the pair is exchangeable; for non-identically distributed summands the conditional expectations differ.
- $\mathbb{E}[X_1 | S] = S/2$ does not mean $X_1 = S/2$: the residual $X_1 - S/2$ is uncorrelated with S but is not zero.

3 Bayesian Posterior Densities

3.1 Conditional Densities from Joint Densities

Proposition 3.1 (Bayes' formula for densities). *Suppose (Θ, X) has joint density $f(\theta, x) = f_{\Theta}(\theta)f_{X|\Theta}(x | \theta)$ with respect to $\lambda \times \lambda$ and let $f_X(x) = \int f(\theta, x) d\theta$ be the marginal, assumed positive. Then*

$$\mathbb{P}(\Theta \in A | X) = \int_A f_{\Theta|X}(\theta | X) d\theta, \quad f_{\Theta|X}(\theta | x) = \frac{f_{X|\Theta}(x | \theta)f_{\Theta}(\theta)}{f_X(x)}.$$

Proof. The right side is a function of X , hence $\sigma(X)$ -measurable. For $G = \{X \in B\}$, Tonelli gives

$$\mathbb{P}(\Theta \in A, X \in B) = \int_B \int_A f(\theta, x) d\theta dx = \int_B \left(\int_A f_{\Theta|X}(\theta | x) d\theta \right) f_X(x) dx = \mathbb{E} \left[\mathbf{1}_B(X) \int_A f_{\Theta|X}(\theta | X) d\theta \right],$$

which is the defining property of conditional probability. $\square$

3.2 Problem: Uniform Prior with a Cauchy Likelihood

Let $0 < a < b$, $\Theta \sim \text{Uniform}(a, b)$, and $X | \Theta = \theta \sim \text{Cauchy}(0, \theta)$:

$$f_{X|\Theta}(x | \theta) = \frac{\theta}{\pi(\theta^2 + x^2)}, \quad f_{\Theta}(\theta) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(\theta).$$

Marginal density. Since $\int \frac{\theta}{\theta^2 + x^2} d\theta = \frac{1}{2} \ln(\theta^2 + x^2)$,

$$f_X(x) = \frac{1}{\pi(b-a)} \int_a^b \frac{\theta}{\theta^2 + x^2} d\theta = \frac{1}{2\pi(b-a)} \ln \left(\frac{b^2 + x^2}{a^2 + x^2} \right),$$

which is positive for every real x because $b > a > 0$.

Posterior density. For $\theta \in [a, b]$,

$$f_{\Theta|X}(\theta | x) = \frac{f_{X|\Theta}(x | \theta)f_{\Theta}(\theta)}{f_X(x)} = \frac{2\theta}{(\theta^2 + x^2) \ln \left(\frac{b^2 + x^2}{a^2 + x^2} \right)}.$$

Check that it is a density. $\int_a^b \frac{2\theta}{\theta^2 + x^2} d\theta = \ln \left(\frac{b^2 + x^2}{a^2 + x^2} \right)$, which cancels the denominator, giving total mass 1. $\square$

Posterior mean. Although the Cauchy likelihood has no mean, the posterior is supported on the bounded interval $[a, b]$, so $\mathbb{E}[\Theta | X = x]$ exists. Writing $L(x) = \ln \frac{b^2 + x^2}{a^2 + x^2}$ and using $\frac{\theta^2}{\theta^2 + x^2} = 1 - \frac{x^2}{\theta^2 + x^2}$, for $x \neq 0$

$$\mathbb{E}[\Theta | X = x] = \frac{2}{L(x)} \int_a^b \frac{\theta^2}{\theta^2 + x^2} d\theta = \frac{2}{L(x)} \left[(b-a) - x \left(\arctan \frac{b}{x} - \arctan \frac{a}{x} \right) \right],$$

and at $x = 0$ it equals $\frac{b-a}{\ln(b/a)}$ (the logarithmic mean of a and b).

Remark (Behaviour of the posterior). • At $x = 0$ the posterior density is proportional to $1/\theta$ on $[a, b]$ (small scales are favoured, since an observation near zero is most likely under a small scale).

- As $|x| \rightarrow \infty$, $\ln \frac{b^2+x^2}{a^2+x^2} \approx \frac{b^2-a^2}{x^2}$, so the posterior tends to $\frac{2\theta}{b^2-a^2}$ on $[a, b]$, proportional to θ (large scales are favoured).
- Only the likelihood shape enters: the posterior is proportional to $\frac{\theta}{\theta^2 + x^2} \mathbf{1}_{[a,b]}(\theta)$, and the normalizing constant is $f_X(x)$ up to a factor.

Note:

- The prior support restricts the posterior: $f_{\Theta|X}(\theta | x) = 0$ for $\theta \notin [a, b]$, however large the likelihood is there.
- Proposition 3.1 needs $f_X(x) > 0$; where the marginal vanishes the formula is undefined (here it never does).
- Do not confuse the density of $X | \Theta = \theta$ (a function of x) with the likelihood (the same expression as a function of θ).

4 Weak Law of Large Numbers and Slutsky's Theorem

4.1 Statements

Write $\bar{X}_m = \frac{1}{m} \sum_{i=1}^m X_i$ for the sample mean (called S_m in the assignment; here S was used for a sum in Section 2).

Definition 4.1 (Convergence in probability). $Z_m \xrightarrow{\mathbb{P}} Z$ if $\mathbb{P}(|Z_m - Z| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Theorem 4.2 (Weak Law of Large Numbers). If $X_1, X_2, \dots$ are i.i.d. with $\mathbb{E}|X_1| < \infty$, then $\bar{X}_m \xrightarrow{\mathbb{P}} \mathbb{E}[X_1]$.

Proof. Finite variance. If $\text{Var}(X_1) = \sigma^2 < \infty$, Chebyshev gives $\mathbb{P}(|\bar{X}_m - \mu| > \varepsilon) \leq \frac{\sigma^2}{m\varepsilon^2} \rightarrow 0$.

General case (truncation). Given $\varepsilon, \delta > 0$ choose K with $\mathbb{E}|X_1 - X_1^{(K)}| < \varepsilon\delta$, where $X_i^{(K)} = X_i \mathbf{1}_{\{|X_i| \leq K\}}$. The averages of the bounded $X_i^{(K)}$ converge in probability to $\mathbb{E}[X_1^{(K)}]$ by the finite-variance case, and $\mathbb{P}(|\bar{X} - \bar{X}^{(K)}| > \delta) \leq \mathbb{E}|X_1 - X_1^{(K)}|/\delta < \varepsilon$ by Markov. Combining with $|\mathbb{E}[X_1] - \mathbb{E}[X_1^{(K)}]| < \varepsilon\delta$ yields $\limsup_m \mathbb{P}(|\bar{X}_m - \mu| > 3\delta) \leq \varepsilon$, and ε is arbitrary. $\square$

Theorem 4.3 (Continuous mapping, constant limits). If $Z_m \xrightarrow{\mathbb{P}} c$ and g is continuous at c , then $g(Z_m) \xrightarrow{\mathbb{P}} g(c)$.

Proof. Given $\varepsilon > 0$ pick $\delta > 0$ with $|z - c| \leq \delta \implies |g(z) - g(c)| \leq \varepsilon$. Then $\{|g(Z_m) - g(c)| > \varepsilon\} \subseteq \{|Z_m - c| > \delta\}$, whose probability tends to 0. $\square$

Theorem 4.4 (Slutsky). *Let $Z_m \xrightarrow{d} Z$ and $W_m \xrightarrow{\mathbb{P}} c$, a constant. Then $Z_m + W_m \xrightarrow{d} Z + c$ and $Z_m W_m \xrightarrow{d} cZ$, and if $c \neq 0$, $Z_m/W_m \xrightarrow{d} Z/c$. In particular, if $Z_m \xrightarrow{\mathbb{P}} a$ and $W_m \xrightarrow{\mathbb{P}} c$, then $Z_m + W_m \xrightarrow{\mathbb{P}} a + c$ and $Z_m W_m \xrightarrow{\mathbb{P}} ac$.*

Proof of the in-probability special case. For sums, $\{|Z_m + W_m - a - c| > \varepsilon\} \subseteq \{|Z_m - a| > \varepsilon/2\} \cup \{|W_m - c| > \varepsilon/2\}$. Products follow from $Z_m W_m - ac = (Z_m - a)W_m + a(W_m - c)$ together with $W_m \xrightarrow{\mathbb{P}} c$. $\square$

4.2 Problem: Consistency of the Sample Variance

Let X_i be i.i.d. with mean μ and variance $\sigma^2 < \infty$, and set

$$\hat{\sigma}_m^2 = \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X}_m)^2.$$

Claim: $\hat{\sigma}_m^2 \xrightarrow{\mathbb{P}} \sigma^2$.

Proof. Expansion. Since $\sum_i (X_i - \bar{X}_m) = 0$,

$$\hat{\sigma}_m^2 = \frac{1}{m} \sum_{i=1}^m X_i^2 - 2\bar{X}_m \cdot \frac{1}{m} \sum_i X_i + \bar{X}_m^2 = \left(\frac{1}{m} \sum_{i=1}^m X_i^2 \right) - \bar{X}_m^2.$$

1. By WLLN, $\bar{X}_m \xrightarrow{\mathbb{P}} \mu$, and by continuous mapping with $g(z) = z^2$, $\bar{X}_m^2 \xrightarrow{\mathbb{P}} \mu^2$.
2. The variables $Y_i = X_i^2$ are i.i.d. and integrable, with $\mathbb{E}[Y_1] = \mathbb{E}[X_1^2] = \sigma^2 + \mu^2$, so WLLN gives $\frac{1}{m} \sum_i X_i^2 \xrightarrow{\mathbb{P}} \sigma^2 + \mu^2$. (Only $\mathbb{E}[Y_1] < \infty$ is needed, not a fourth moment.)
3. By Slutsky (in-probability form),

$$\hat{\sigma}_m^2 \xrightarrow{\mathbb{P}} (\sigma^2 + \mu^2) - \mu^2 = \sigma^2. \quad \square$$

Corollary 4.5 (Unbiased sample variance and the t -statistic). *Let $s_m^2 = \frac{m}{m-1} \hat{\sigma}_m^2$. Then $s_m^2 \xrightarrow{\mathbb{P}} \sigma^2$, and if $\sigma^2 > 0$,*

$$\frac{\sqrt{m}(\bar{X}_m - \mu)}{s_m} \xrightarrow{d} N(0, 1).$$

Proof. $\frac{m}{m-1} \rightarrow 1$ and $\hat{\sigma}_m^2 \xrightarrow{\mathbb{P}} \sigma^2$, so $s_m^2 \xrightarrow{\mathbb{P}} \sigma^2$ and $s_m \xrightarrow{\mathbb{P}} \sigma$ by continuous mapping. By the CLT, $\sqrt{m}(\bar{X}_m - \mu)/\sigma \xrightarrow{d} N(0, 1)$, and Slutsky's theorem (division by $s_m/\sigma \xrightarrow{\mathbb{P}} 1$) gives the claim. $\square$

Remark. $\hat{\sigma}_m^2$ is consistent but biased: $\mathbb{E}[\hat{\sigma}_m^2] = \mathbb{E}[X_1^2] - \mathbb{E}[\bar{X}_m^2] = (\sigma^2 + \mu^2) - (\sigma^2/m + \mu^2) = \frac{m-1}{m} \sigma^2$. Consistency (convergence in probability) does not require unbiasedness, and the bias vanishes at rate $1/m$.

Note:

- Slutsky's theorem in its distributional form needs one of the two sequences to converge to a *constant*; two sequences converging in distribution to non-degenerate limits cannot be added or multiplied limit-wise without joint information.

- The WLLN for X_i^2 needs $\mathbb{E}[X_1^2] < \infty$, i.e. finite variance of X_i , which is assumed here; without it the sample variance need not be consistent.
- Convergence in probability is weaker than a.s. convergence (SLLN); the WLLN concerns the law of $\bar{X}_m$ at each fixed m , not the sample path.

References

- [1] S. Basu, T. Butler, D. Estep, and N. Panda, *A Ramble Through Probability: How I Learned to Stop Worrying and Love Measure Theory*, Computational Science & Engineering, vol. 29, SIAM, Philadelphia, 2024.